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Ant colony prediction by using sectorized diurnal mobility model for handover management in PCS networks

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Abstract Recently, mobile phones are extremely used in lifestyle. Historical records of mobile users (MUs) play an important role in predicting future movements of new visitors of the underlying registration area. Handover (handoff) is one of important quality of service (QoS) parameter that affects the continuity of the call when MUs move from a cell to its neighbors in the same registration area (RA). In this paper, a novel ant based Algorithm, has been introduced, which is called Ant Prediction Algorithm (APA). The main target of APA is to reduce handover impact on the performance of personal communication service (PCS) networks. To accomplish such aim, APA tries to minimize the number of dropped calls by predicting the long-term movement of MUs based on the Sectored Diurnal Mobility Model (SDMM). APA consists of two Parts, namely; (i) the Ant Prediction Engine (APE), which relies on the movement history of the other MUs to predict the future movement of the considered MU, and (ii) the SDMM design, which predicts the exact future sector and cell of the considered MU. Simulations have been presented to validate the proposed scheme in terms of prediction accuracy and handoff blocking probability.

Keywords Personal communications networks · Handover schemes · Prediction theory · Ant colony

1 Introduction

In personal communication service (PCS) networks, two ordinarily used standards, the EIA/TIA Interim commonplace in North America and the Global System for Mobile Communications in Europe (GSM). Both standards partition their coverage areas in a variety of location areas (LA), every consisting of a bunch of cells. With the growing request for brand new services and utilities, PCS networks ought to support calls with varied traffic options and varied Qos guarantees. Additionally, very different completely wireless networks and technologies stay really which will satisfy different desires and needs of MUs. In the procedure PCS networks, mobility management deals with the tracking, storage, maintenance, and retrieval of mobile location information. In real-life situation, Mobile Users (MUs) principally notice their destinations a priori, their movement isn't fully random. Hence, valuable information can be concluded from the previously registered paths of MUs that were passed through the underlying network service area (SA).

In PCS Network, a SA is divided into a variety of registration areas (RAs), each RA is subdivided into various cells. A cell is that the communication space wrapped by one Base station (BS). BSs within the same RA are connected to at least one Mobile Switching Center (MSC) that may be a dedicated telephone switch for PCS applications. BSs are usually static unit and are connected through land links to the MSC serving their RA [1]. Each SA is served by a global database called Home Location Register (HLR), which includes the records of all users' data

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including their current location, billing information, as well as their allowed services. On the other hand, another database called Visitor Location Register (VLR), which transfers information from the HLR around current MU that are currently visiting its Location area (LA). Hence, every LA has one VLR to service it, and every VLR is used to supervise one LA [2, 3].

Swarm Intelligence models are computing calculation models utilized for settling complex dispersed streamlining issues. The fundamental standard of Swarm Knowledge principally focusses on "Probabilistic-based search Algorithms" [4–6]. Ant Colony Optimization (ACO) is one of the models used to solve several optimization problems. The key issues in ACO are the movement of ants (heuristic) and the pheromone update/control [7–9].

Many researches used Ant System to find best solutions. The ant Colony optimization Metaheuristic relies on generic drawback illustration and the definition of the ant's behavior [10]. One of the problems that can be solved by using the ant system is a prediction of user movement in next steps. It allows highly exact prediction of user's displacement even without any information. Ants' movements are influenced by social behavior. Ants prefer to choose the trail where there are more pheromones [11, 12].

In [13] shows a reinforced proceeding for Mobile IPv6 path optimization, which utilizes the basics of shared secret Token and one-time password with a verification of the mobile user through an immediate telecommunication [data packet service]. In [14] introduces two steps of vertical handover decision algorithm based on a Dynamic Weight Compensation (DWC). In [15] a handoff optimization algorithm relies on the MU's mobility status by using handover signaling cost analysis. In [16] discuss the drawbacks of a handoff management in heterogeneous cloud small cell network (HCSNet).

In this paper, we proposed APA that introduce the best predict path of mobile user [voice call service] in pcs network to reduce horizontal handover which happen when mobile user mover from one cell to another cell which have same technology (within the same pcs network) by predicting the next movement of MU mobility. This Algorithm depends on Ant system, which is similar to human behavior. APA is used mainly to predict the next location of the MU who has insufficient history in the current location. It uses the history of neighbors who are in the same region. The proposed APA algorithm is divided into two Parts, namely, (i) the Ant Prediction Engine (APE), which is based on the history of the other MUs to predict the next location of the MU of interest, and (ii) Sectored Diurnal Mobility Model (SDMM), which is a combination of two mobility models (sectorized and diurnal mobility models).

The remainder of the paper is organized as follows; Related works in this field in Sect. 2. The proposed Ant Prediction Algorithm (APA) by using the Ant Prediction Engine (APE) and sectorized diurnal mobility model (SDMM) Algorithm is explained in Sect. 3. Experimental results are given in Sect. 4. Finally, Sect. 5 concludes the paper.

2 Related work

Recently, mobile phones are widely applied in the modern community. People create phone calls at home, at work, and while traveling. With the quick growth of mobile phones and data aggregation techniques, it is currently possible to register all phone calls created during a town from operators in the form of phone records. Those records typically contain caller and caller IDs, date, time, call length and the base station from where the telephone call was initialized.

In [17], the authors suggest a Hybrid Mobility Prediction (HMP) strategy, which gives adaptable utilization of the limited resources of PCS networks. It consolidates prove from three distinct indicators, which are; Probabilistic Predictor (PP), Group based Predictor (GP), and Spatial Predictor (SP). PP depends on Bayes theorem; GP uses Ant Colony Optimization (ACO), while SP tries to distinguish the topological design of the present enrollment zone to improve the expectation procedure. On the other hand, Lu et al. [18] provides the idea concerning ant colony optimizing algorithm that supported load equalization. Claes and Holvoet [19] and Claes et al. [20] characterize the utilization of ants as a method for gathering the knowledge spread across a variety of margin computers. In [21], a cell-sector numbering scheme has been used by dividing every cell to six sectors such as the cell (a) is to be (a1, a2, a3, a4, a5, a6). This has a good impact in predicting the next handover. In [22] the probability of user movement in the six surrounding cells has been estimated, but two of them that have the higher probability are considered.

The authors in [23] proposed a new prediction model that is based on people movement by monitoring the movement history for every MU through the mobile telecom service. The objective of the work in [24] is to predict user's displacement based on MU's behavior using ant colony optimization (ACO). In [25] a Dynamic Clustering based Prediction (DCP) of MU's movements has been introduced. DCP is employed in locating MU's quality patterns from collections of recorded trajectories. Then, these patterns are used for the prediction of movements similarly as dynamically allocating network resources. In [26] an enhanced algorithm is proposed which utilizes both ACO as well as a neuro-fuzzy framework.

Authors of [27] propose a link prediction algorithm based on ACO. By exploiting the swarm intelligence, the algorithm utilizes artificial ants to go on a logical diagram. Pheromone and heuristic information are allowed in the edges of the logical diagram. In [28] a model for finding the ideal route based on the integration of traffic data from various sources has been proposed, while [29] proposed another idea in which MUs can keep up a database (archive table) of available networks characterized by their quality and introduced a new vertical handover mechanism that fulfills more measurements and offers better QoS using ant colony approach.

In [30] another Mixed Movement Predictor (MMP), which diminishes the impact of handover issues through a perfect expectation without bounds developments for arrange MUs, has been proposed. This system depends upon two structures to cover all probabilities. The first is Path Matching Algorithm (PMA), which is used when a MU has no history. It is considered of symmetry between the present route for the MU enrolled utilizing GPS/DR and past tracks have a place with a comparative way that used by inhabitants in this area. The second structure is the Fuzzy Prediction Engine (FPE), which is utilized when the MU has a history. It relies on upon six variables, specifically; GPS, Distance, Famous Places, Time, Date, and User Profile to anticipate the future client portability.

In [31] a new energy optimization strategy for the ant colony algorithm is proposed. The proposed strategy uses the separation between the nodes and the residual energy of the nodes in order to choose the next hop and assigns certain weights to the energy consumption and the pheromone concentration.

A handoff optimization scheme depends on the cell id data which aims to improve the handoff parameters and develops the efficiency of operation [32]. The Ant Colony Algorithm depends on mobility robustness optimization in LTE/LTE-A Femtocell networks. It utilizes mutual of the cost task [33].

In [34] introduce a useful description of the Fog radio access network (FRAN) structure for the 5G network, and it focuses on the resource optimization, interference mitigation, and mobility management in FRAN. In [35] provide a logical structure for network slicing depend on 5G systems and introduce a diagram for managing mobility between various access networks.

In this paper, we propose a new Algorithm to reduce the handover problem by predicting the future movement of network MUs. we focus on inter BSC or it named intra MSC handover, which is the handover between cells controlled by different BSCs and it controlled by the same MSC in the PCS network. This Algorithm is called Ant Prediction Algorithm (APA), which depends on neighbors' history. APA has the ability to predict the future user mobility based on Ant Prediction Engine (APE) and a Sectored Diurnal Mobility Model (SDMM).

3 The proposed ant prediction algorithm (APA)

Through this section the proposed Ant Prediction Algorithm (APA) will be explained in details. APA is specialized in predicting the next movement of new MUs or those users who have no sufficient history in the current location. As a prediction system, there are two stages to predict next movement of a MU. The first stage is Ant Prediction Engine (APE) which uses an ant system to predict the future movements of MUs in the network. The second stage is Sectored Diurnal Mobility Model (SDMM), which is a combination of two mobility models, namely; sectorized mobility model and diurnal mobility model. The proposed prediction system is illustrated in Fig. 1.

As illustrated in Fig. 1, mobility database (MDB) stores the complete profiles of MUs. It consists of two parts. The first is the permanent profile database (PPDB), which contains movement history for every MU and records the cells crossed by the MU during having a call. The second part of MDB is the temporary profile database (TPDB), which record 3 elements (speed, direction, position) for the MU when having a call, these elements that vary with the time. The steps of the proposed prediction can be summarized as follow:

Step 1 When MU receives a call.
Step 2 Define the location of this mobile on a map.
Step 3 Collect information about this MU from Mobility database (MDB).
Step 4 Information from MU's profile, history will be derived if that MU is available in this RA before or it is the first time.
Step 5 If the MU is a visitor (rarely found in this region), Ant Prediction Engine (APE) will be used. But, if the MU has a history, other prediction technique is suitable for applying in this case.
Step 6 Ant Prediction Engine (APE) gives the first prediction of the future cell.
Step 7 To raise the accuracy of prediction system, Sectored diurnal mobility model is used, which defines the MU at the future by scattered cells and predicted future sector with the diurnal model that defines MU by getting the angle of the movement direction.
Step 8 The outcome of Sectored Diurnal Mobility Model (SDMM) is the exact future sector and cell of the considered MU.

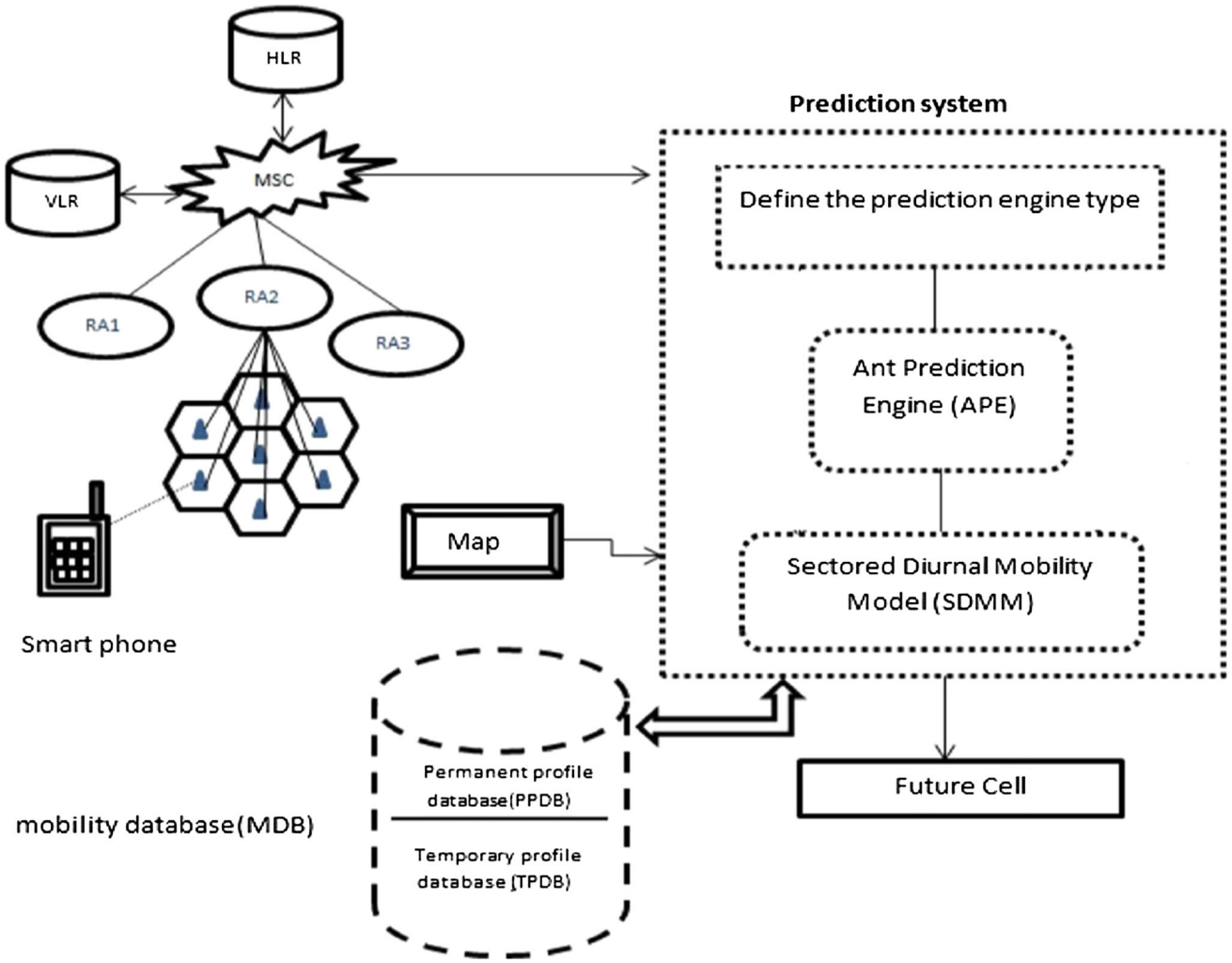


Fig. 1 Proposed prediction algorithm

3.1 Ant prediction engine (APE)

Ant Prediction Engine is a method using the ant system to predict the future movements of mobile users in the network. A new mobile user in the cell who does not have any history or the user who does not have a sufficient history will choose the path that is used by his neighbors. There is a great similarity between the ants and the mobile users in the network.

Service area is a group of cells and registration area is a part of it. In Ant Prediction Engine (APE), registration area looks like a flower which has the center (cf) and six of Floral parts (f1, f2, f3, f4, f5, f6). Ants represent mobile users; pheromone that is placed by the ant represents user data that are saved in the network history. When a mobile user moves from cell F4 to F2 via the center of a flower as shown in Fig. 2, the Center cell (cf) will record this displacement by adding a new input to its history. The Ant chooses the road that the most number of other ants passing this road, the more the amount of pheromone. There are various parameter employed in APE as discussed in Table 1.

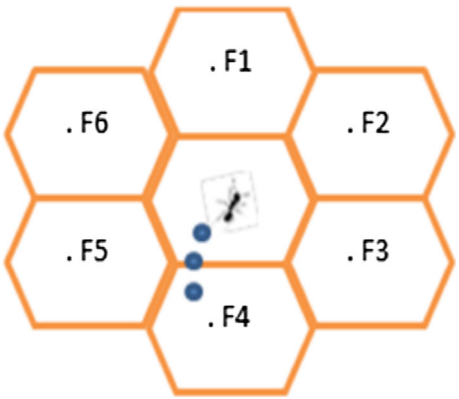


Fig. 2 Ant movement prediction (AMP)

Table 1 Various parameter employed in AMP

N	The number of ants which entered the cells at time t
M	The number of ants which entered the cells at the time (t + 1)
$\boldsymbol{\mu}$	Evaporation rate
K	The total number of ants
Q	The amount of pheromone by one ANT
$p_{x(\mathrm{f4}\rightarrow\mathrm{f2,t})}$	The probability of pheromone from X when it move from cell 4 to cell 2
$X_{-displacement}(\mathrm{f4}\rightarrow\mathrm{f2,t})$	The displacement by one ant
$\Delta D \cdot P(t)$	The density of pheromone
P.P	Path probability

As illustrated in Fig. 2, Let the ant x At the time (t) when MU_i move from f4 to f2, the probability of pheromone from cell 4 to cell 2 can be calculated using (1).

$$p_{x(\mathrm{f4}\rightarrow\mathrm{f2,t})} = X_{-displacement}(\mathrm{f4}\rightarrow\mathrm{f2,t}) \Big/ \sum_{m=1}^{k} X_{-displacement}(\mathrm{f4}\rightarrow\mathrm{f}m,\mathrm{t}) \tag{1}$$

It is known that; ants choose the path containing the largest quantity of pheromone, which is called pheromone density. If one ant puts the amount of pheromone Q at time t, after passing n ants, the density of pheromone can be calculated using (2).

$$\Delta D \cdot P(t) = N * Q \tag{2}$$

We assume that all ants have the same quantity of pheromone, So Q = 1 and density of pheromone will depend on the number of ants, consider (3).

$$\Delta D \cdot P(t) = N \tag{3}$$

However, after a certain period to put pheromone on the road, the evaporation process occurs. This process is the same when MU's data becomes old and useless, so it should be deleted from history.

Let $\boldsymbol{\mu}$ to be the evaporation rate (between time t and (t + 1)) and M is the number of new ants, which enters into cells at the time (t + 1). Hence, the density of pheromone at time t + 1 can be calculated using (4).

$$\Delta D \cdot P(t+1) = (1-\boldsymbol{\mu}) * \Delta D \cdot P(t) + M \tag{4}$$

The choice of the path on the density of pheromones, therefore path probability (P.P) that the ant is chosen is proportional to the density of pheromone as illustrated in (5).

$$\mathrm{P.P} = \alpha \Delta D \cdot P(t+1) \tag{5}$$

Hence, path probability can be estimated by (6).

$$\mathrm{P.P} = \lambda * \Delta D \cdot P(t+1) \tag{6}$$

where P.P is path probability, λ is fixed proportionality.

As illustrated in Table 2, let $\mu = 0.3, \lambda = 0.1$. Considering Fig. 2 again, by applying previous equations the future cell that the ant will choose is cell 2 as illustrated in Fig. 3. In this example, the ant depends on the number of ants (N) that entered the cells at time t, the evaporation rate (μ) and the number of ants (M) that entered the cells at the time (t + 1).

Table 2 Mobile user's table

Mobile ID	Source cell	Destination cell	Date	Time
24578955	F4	F2	Monday	1:05
24587954	F4	F2	Monday	12:40
2457666	F4	F2	Monday	1:15

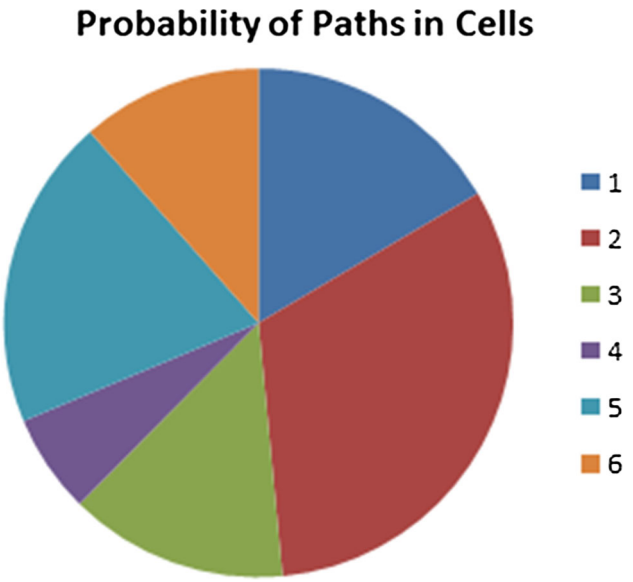


Fig. 3 The probability of paths in cells

The Ant Prediction Engine will use an ant system that depends on MUs in neighboring cells; native MUs used the paths, which connect their homes, workplaces and famous buildings. Therefore, the new MU that comes to the center of the flower (Cf) which is the centers of neighbor cells will choose the same path used by users who come from the same source cells, time and date as shown in Fig. 4.

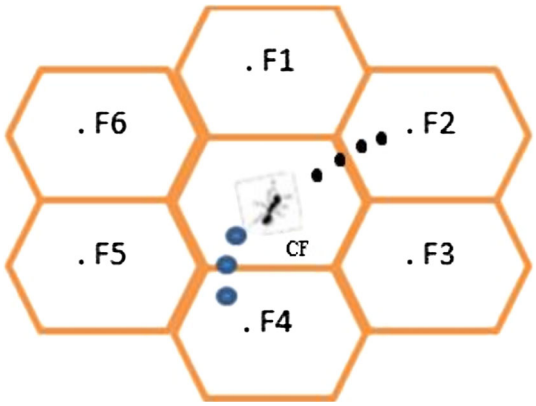


Fig. 4 Registration area has CF

Each cell in registration area is a center of the flower according to other neighbors' cells, so every CF has movement user's table as the example (Table 2).

When a MU moves from the same source cell to another cell, it will move through the center of the flower, which will check if this MU has a history, or not. Ant Prediction Engine is invoked only if MU has no or insufficient history and it depends on neighbors' histories which have the exact same source cell and approximate date and time.

3.1.1 Ant prediction algorithm (APA)

The proposed APA relies on an ant system. In this algorithm, the MU is searching for neighbor history, APA used the data, which is stored in MDB and updating the MUs' data accordingly. Details of APA are illustrated in the next paragraphs.

1. When the first ant moves from the nest to search the food, it takes any path randomly the same as the new mobile user without any information about this path or the user who carries out unusual movement.
2. The ant chooses the path with a high of pheromones quantity the same as searching for the number of people who have the same feature (source, time and date) to know the path with a high density of mobile users.
3. The evaporation of pheromones similar to the updating of the mobile user history records by deleting old data.
4. Finally, the ant discovers the path with the highest quantity of pheromones the same as APE algorithms when finding the path that mobile user pass it.

3.2 Sectored diurnal mobility algorithm (SDMA)

In real-life scenario, MUs mostly realize their destinations a priori. Hence, their movement is not completely random. Actually, it is direction-based and depends on the current location and the final movement destination.

In Sectored Diurnal Mobility Algorithm (SDMA), the cell (A) in Fig. 5, will be divided into six sectors by using 3 axes. Each axe will make angel 60° to every sector, so if a MU is in sector a3 with respect to the closet directions leading to the center of neighboring cells, it has two angles for movement Q1, Q2 as depicted in Fig. 4. Accordingly, the probability for mobility can be calculated using (7) and (8).

$$P1 = (1 - Q1/60) \tag{7}$$

$$P2 = (1 - Q2/60) \tag{8}$$

The movement of MU is stored in the mobility history base sector by sector as (a2 → b3 → f5 → h4 → g2 → k5) instead of cell-by-cell. The sectored diurnal mobility structure is to have three regions as illustrated in Fig. 6.

- The region where inner or close to the base station, the probability of handoff is negligible (No handoff).
- The region where is in middle of the cell, the probability of handoff is 50% (low handoff).
- The region where is in outer of the cell, the probability of handoff is high (high handoff).

The method for estimating MU's movement direction is to get the average of angles for the previous movement in this cell and the next direction will be the summing of the angles divide by the number of them as depicted in (9) and shown in Fig. 7.

$$\text{The predicting of direction (PQ)} = (Q1 + Q2 + Q3 + Q4 + Q5)/N \tag{9}$$

As shown Sectored Diurnal Mobility Algorithm (SDMA) which stores the locations of the MU on a sector-by-sector basis instead of a cell-by-cell basis. From path of the movements of the MU, we can distinguish handover

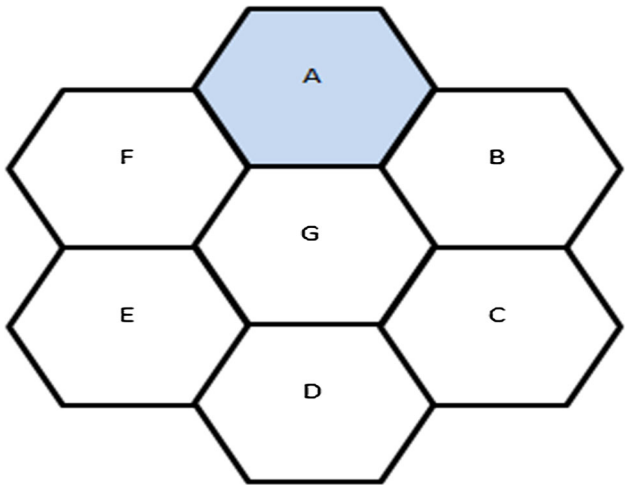


Fig. 5 Registration area has 7 cells

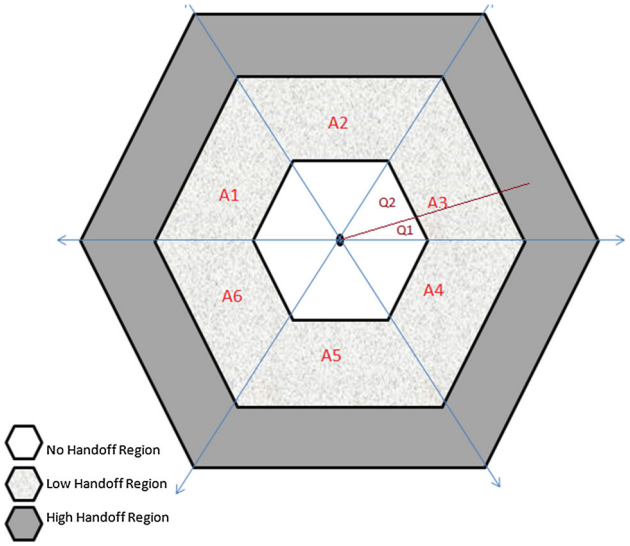


Fig. 6 Sectored diurnal mobility model (SDMM)

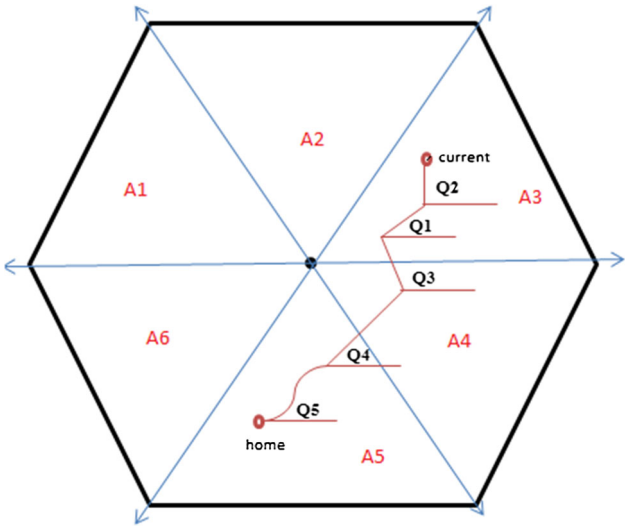


Fig. 7 Predict of direction (PQ)

points as the MU moves from cell to another cell, the cell-sector numbering scheme is used as shown in Fig. 8.

Definition 1 Cell-sector numbering scheme.

It a numbering methodology that divides every cell has six sectors. If Current Sector is a (I) & Predicted cell is B then,

$$\text{Predicted sector} \longrightarrow \begin{cases} \text{B (I + 3)} & \text{when I = < 3} \\ \text{B (I – 3)} & \text{when I > 3} \end{cases} \quad (10)$$

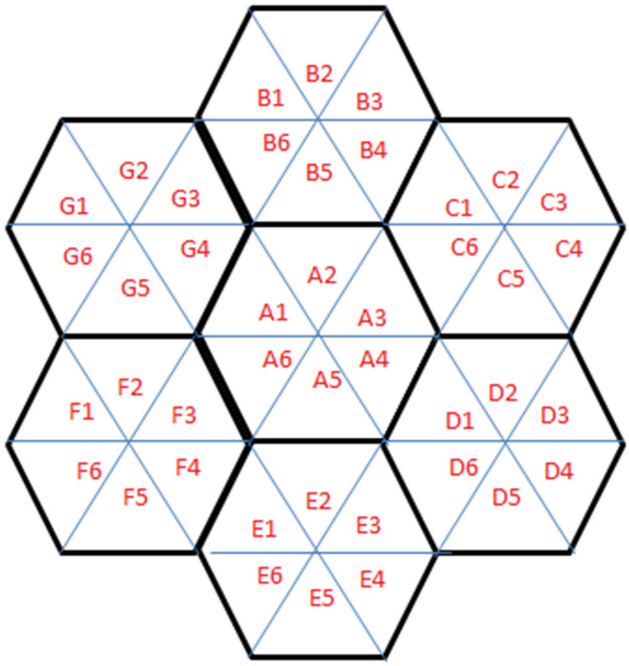


Fig. 8 The cell-sector numbering scheme

4 Experimental results

The simulation has been done in ns-2 to determine the movement of MUs using APA, which based on the Ant Prediction Engine (APE) and a Sectored Diurnal Mobility Model (SDMM). The simulation was done according to MUs = {100, 200, 300, 400, 500, …, 1000}, the simulated service area contains 40 base stations with a cell radius of 3 km, speed range for MTs from [40–60] miles/h.

4.1 Prediction accuracy

The prediction process starts from the source location of the cell until the current location of the cell. Only one MU is considered to trial the various prediction algorithms. The first location of such MU is randomly chosen in the considered registration area of 36 cells. Various prediction algorithms are then applied to predict its next cell. Those algorithms are; Dynamic Clustering based Prediction (DCP) [25], History Based Prediction (HBP) [23], Mixed Movement Predictor (MMP) [30] and Hybrid Mobility Prediction (HMP) [17]. Then, performance evaluation is performed by calculating the prediction accuracy (p1) that defined by the ratio of the number of handover failure to the total number of predictions as expressed in (11).

Results are illustrated in Fig. 9. Prediction Accuracy (P2) is the ratio between the number of correct predictions

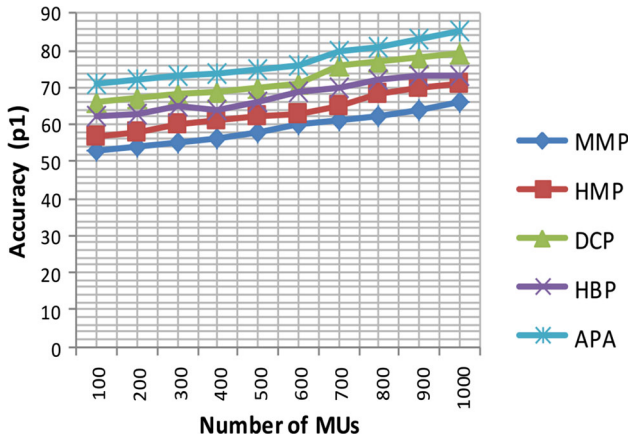


Fig. 9 Prediction accuracy (p1)

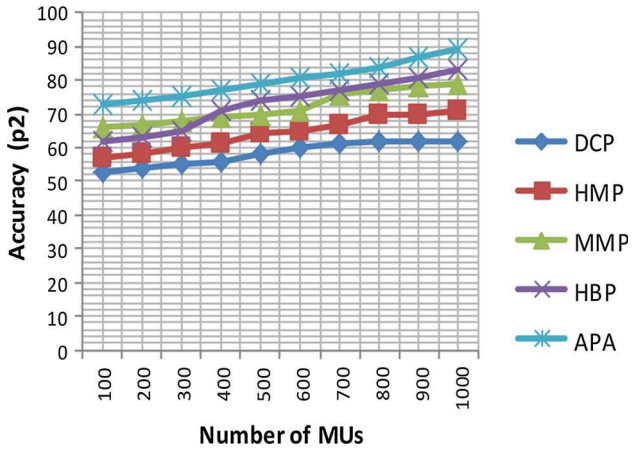


Fig. 10 Prediction accuracy (p2)

and the number of all predictions which can be calculated by (12). Results are depicted in Fig. 10. Finally, Prediction Accuracy (P3) is the ratio between the number of MUs executed handoff (without history) and the number of all

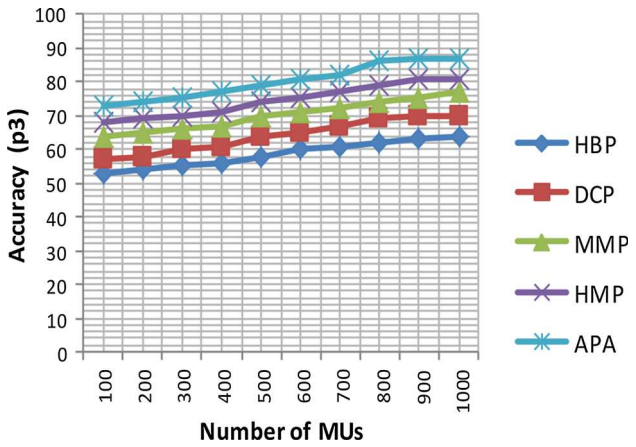


Fig. 11 Prediction accuracy (p3)

MUs executed handoff which can be calculated by (13). Results are depicted in Fig. 11.

$$\mathrm{P1} = \frac{\textit{Number of handover}}{\textit{Total number of predictions}} \tag{11}$$

$$\mathrm{P2} = \frac{\textit{Number of correct prediction}}{\textit{Total number of predictions}} \tag{12}$$

$$\mathrm{P3} = \frac{\textit{Number of MTs executed handoff (without history)}}{\textit{Total number of executed handoff}} \tag{13}$$

Figure 9 illustrates the prediction accuracy against the number of handover failures according to the total number of predictions for APA and other competitors, DCP and APA give the best accuracy (P1) as the number of handover failure is less than other competitors. On the other hand, Fig. 10 illustrates the prediction accuracy (P2), MMP and APA give the best result for correct prediction results. Figure 11 shows the prediction accuracy (p3). It can be concluded that APA is the best algorithm to predict next movements of MUs (without history). APA outperforms all other competitors in all prediction accuracy test as it achieve low number of handover failure, low handover occurrences and give a high correct predictions results.

4.2 Handoff blocking probability (HBP)

Generally, HBP estimates the probability that a call in progress is broken because of handoff (cell change generally due to MU displacements). There are three types of reservation; the first is without reservation (WR), which does not reserve resources for handoffs. The second is Static reservation (SR), which permanently reserves 10% of the cell bandwidth capacity for handoffs. The third is Dynamic reservation (DR), which dynamically reserves resources for a mobile user in adjacent cells. We compare the previous three reservations with our proposed Ant Prediction Algorithm (APA), which depends on neighbors' history to predict future user mobility based on Sectored Diurnal Mobility Model (SDMM). It reserves resources only in predicting future cells using our prediction algorithms. As shown in Fig. 12, simulation results show that our prediction Algorithm APA produces less handoff blocking, which could be attributed to resources reservation in the predicted cells.

4.3 Performance according to time to trigger and hysteresis

Hysteresis and time to trigger are the most important handoff control parameters, which is used to avoid unnecessary handoff. We compare the APA with a handoff self-optimization mechanism (SOM) that introduced in

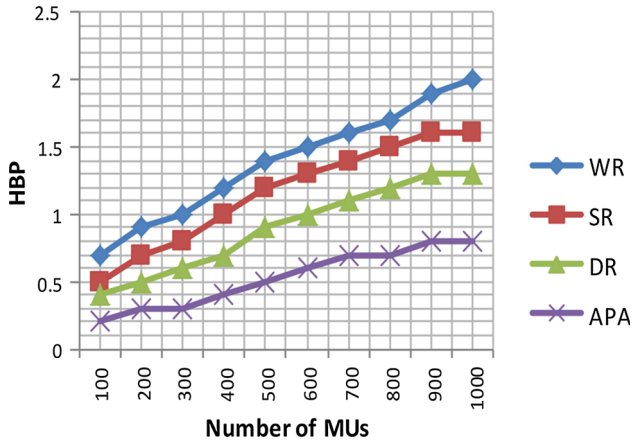


Fig. 12 Handoff blocking probability (HBP)

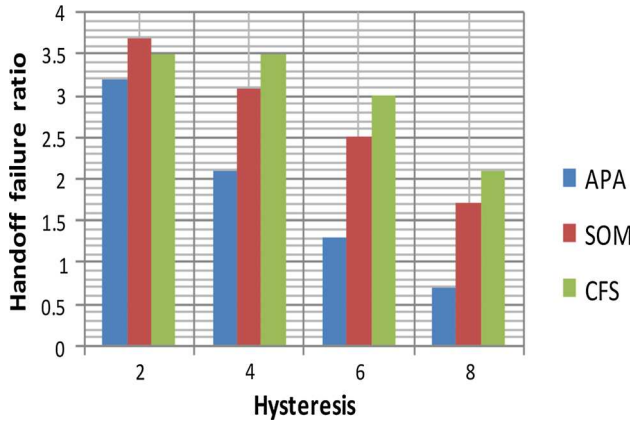


Fig. 13 Handover failure against hysteresis at time to trigger = 100 ms

[36] and the proposed cost function scheme (CFS) that introduced in [37]. As the hysteresis increases in Fig. 13, handover failure decreases because the large amount of

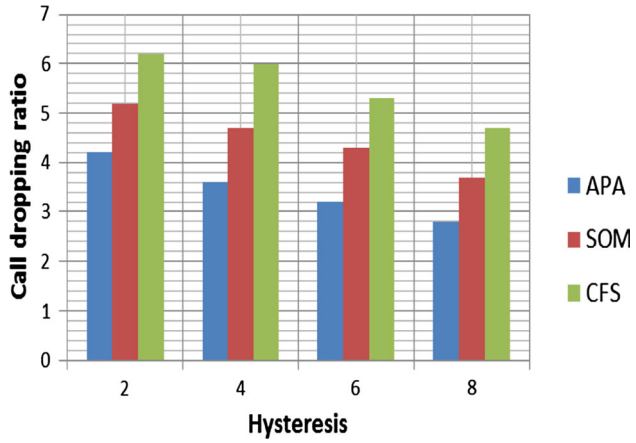


Fig. 14 Call dropping ratio against hysteresis at time to trigger = 100 ms

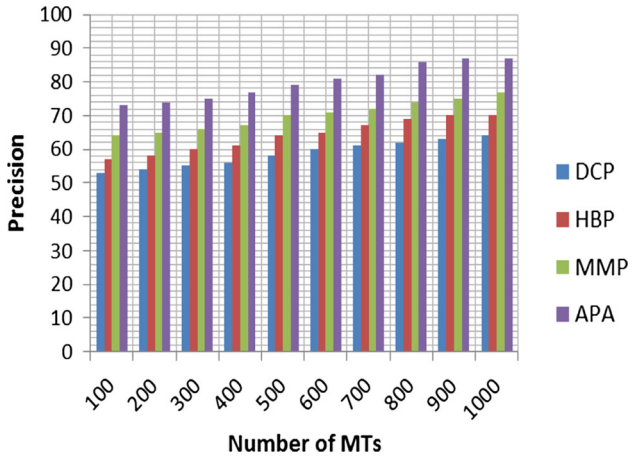


Fig. 15 Ant prediction algorithm (APA) compared to various algorithms

initiated handoff, the small hysteresis cause handover failure ratio. APA gives the minimum value of handoff failure ratio. In Fig. 14, the call dropping ratio decreases as the result of increasing the threshold of hysteresis, MU equipment tends to stay in the serving cell instead of the handoff of target cell which may have a better signal. APA gives the minimum results against other methods.

4.4 Performance according to different prediction algorithms

The actual scale to see the performance of the proposed Ant Prediction Algorithm (APA) is to compare this algorithm with a number of other prediction algorithms. There are many algorithms that expects or the movement of the vehicle or the mobile user in several ways. It has been selected four of them to compare to the proposed APA. The three algorithms are chosen according to the number of metrics to compare with our proposed algorithms and more experiments were done to ensure the precision of them as shown in Fig. 15. The proposed APA give the best results as it focus on mobile users (with no history) and it rely on sectorized cells, so it give the highest prediction results.

5 Conclusions

In this paper, a new efficient Algorithm for solving the handover problem in pcs networks by predicting the future movement of mobile users has been introduced. This Algorithm is called Ant Prediction Algorithm (APA), it is used when the MU is a new one or has insufficient history. It depends on neighbors' history to predict the future MU's mobility. The advantage of this Algorithm is the rapidly reaction with MUs who have unpredictable movements to

predict their future movements. Experimental results have shown that the proposed algorithm introduces good performance as it outperforms the other competitors.

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